TOPIC 1 Linear Equations and Inequalities

Name _____ Date _____

1. What value of x makes $2(x + 3) = x - 4$ true?

A. **−10**

B. −7

C. 2

D. No solution

$2x + 6 = x - 4$, so $x = -10$.

2. How many solutions does $3x + 1 = 3x - 5$ have?

F. One solution, $x = 0$

G. **No solution**

H. Infinite solutions

J. One solution, $x = -2$

Same x-term, different constants: $1 \neq -5$, never true.

3. What are the slope and y-intercept of $y = -2x + 7$?

A. slope 7, (0, −2)

B. slope 2, (0, 7)

C. **slope −2, (0, 7)**

D. slope −2, (7, 0)

In $y = mx + b$, $m = -2$ and $b = 7$.

4. What is the x-intercept of $y = 2x - 6$?

F. (−6, 0)

G. (0, −6)

H. (−3, 0)

J. **(3, 0)**

Set $y = 0$: $0 = 2x - 6$, $x = 3$.